When Heat and Smoke Converge: Mortality and Compound Climate Exposure in the Western United States

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Abstract

Climate change is increasing the frequency and intensity of extreme heat and wildfire activity, exposing populations in the Western United States to multiple environmental hazards simultaneously. While existing research has established that both extreme heat and wildfire smoke independently increase mortality risk, less is known about the health consequences of their co-occurrence. This paper examines whether days when these hazards co-occur impose an additional mortality burden beyond their individual effects. Using county-level panel data from eleven Western U.S. states between 2006 and 2020, the paper constructs a measure of compound heat–smoke exposure defined as the annual number of days on which extreme heat and wildfire smoke occur simultaneously. Then, it estimates its effect on annual all-cause mortality rates while holding overall exposure to heat and smoke constant. By treating co-occurrence as the exposure of interest, the analysis isolates the incremental risk associated with experiencing these hazards jointly rather than separately. The findings suggest that compound exposure is associated with higher mortality rates, but the difference is not statistically significant, suggesting no relationship between the two. While we expected that simultaneous heat and smoke create compounded health stressors, potentially exceeding the sum of their independent effects, the results do not provide any significant reason to reject the null hypothesis. Nevertheless, we suggest further analysis of the topic using broader data, as it is important to conceptualize climate risks not as isolated events but as interacting systems.

1 Introduction

Climate change has increased the likelihood of wildfires and extreme heat. The frequency of active wildfires has risen, with the largest increase in wildfire behavior in temperate conifer forests of the Western U.S. (Cunningham et al., 2024). The duration of fire weather seasons has increased across one-quarter of Earth's vegetated surface (NASA, 2015). The number of extremely hot days, those with temperatures above 95°F (35°C), is expected to rise as a result of climate change in the U.S. (EPA, 2025). While they affect populations worldwide, the western U.S. experiences substantial exposure to both wildfire smoke and extreme heat.

Human populations are increasingly exposed to prolonged wildfire activity and high temperatures, both of which pose serious health risks. Extreme heat can increase dehydration, cardiovascular strain, heat exhaustion, and heat stroke, while exposure to wildfire smoke is associated with respiratory and cardiovascular illness (WHO, 2023, 2024). Moreover, wildfire smoke fine particles, $PM_{2.5}$, are associated with premature death (WHO, 2023). These risks may be particularly severe when extreme heat and wildfire smoke occur simultaneously, as concurrent exposure could impose compounded physiological stress. When these two activities are more apparent and frequent in the western U.S., they would pose an increasingly important public health threat.

This paper estimates the causal effect of annual compound exposure to extreme heat and wildfire smoke on county-level mortality across eleven Western U.S. states from 2006 to 2020. The central question is: What is the causal effect of the annual number of days in which extreme heat and wildfire smoke occur simultaneously ("compound days") on a county's annual mortality rate, holding constant total heat exposure and total smoke exposure? Based on many existing articles and studies, we expect that the greater the annual number of compound heat-and-smoke days, the higher the mortality rate in counties in the Western U.S., given that simultaneous exposure to extreme heat and wildfire smoke may impose compounded physiological stress beyond what either hazard produces independently. Nevertheless, the paper examines the association empirically, without posing directional assumptions in the estimation.

- H_0 (Null): There is no association between the annual number of compound heat-and-smoke days and the mortality rate in counties in the Western U.S.
- H_1 (Alternative): There is an association between the annual number of compound heat-and-smoke days and the mortality rate in counties in the Western U.S.

By defining compound heat–smoke days as the treatment itself, the estimand captures the incremental mortality risk attributable specifically to the co-occurrence of heat and smoke on the same day, beyond the harm of either heat or smoke alone. Identifying this incremental harm is critical for climate adaptation planning: if mortality risk increases disproportionately when heat and smoke occur on the same day, then targeted interventions, such as coordinated cooling and air filtration during forecasted co-exposure days, will have value beyond generic smoke or heat mitigation strategies.

2 Literature Review

There have been studies at the intersection of three rapidly evolving fields: the atmospheric science of extreme events, the epidemiology of air pollution and heat, and the emerging literature on compound climate hazards.

2.1 The Atmospheric Science of Extreme Events

Exposures to wildfire and heat are not static background conditions; they are changing, and they are changing fast. Cunningham et al. (2024) show in *Nature Ecology & Evolution* that the frequency of extreme wildfires has more than doubled since the early 2000s, with the sharpest increases concentrated in temperate conifer forests of the western United States. Burke et al. (2021) sharpen this picture by estimating that the fraction of the U.S. population exposed to smoke-polluted air has grown dramatically since the mid-2000s, with most of the increase concentrated in the West. Anderegg et al. (2022) add a forward-looking perspective, arguing that climate stress, insect outbreaks, and fire risk are likely to reinforce one another in western forests for decades to come.

It means we cannot treat compound heat–smoke days as a rare shock. They are becoming a fixture of summers in states like California, Oregon, and Washington, and the counties in our panel have experienced very different trajectories of exposure over the 2006–2020 window. That heterogeneity in exposure trends across counties, combined with shared calendar-year shocks, is precisely what makes a continuous-treatment difference-in-differences design feasible here.

On the heat side, Ebi et al. (2021) review evidence in *The Lancet* that heatwaves are becoming longer, more intense, and more frequent across nearly every inhabited continent, and the U.S. EPA (2025) reports that days above 95°F are projected to continue rising sharply in the American West. We take these broad trends as given and ask a narrower question: conditional on how often heat and smoke days occur separately, what happens specifically when they coincide?

2.2 The Epidemiology of Air Pollution and Heat

It would be easy to assume that $PM_{2.5}$ from wildfires behaves like any other fine particulate matter, but the literature suggests otherwise. Aguilera et al. (2021), in a careful study published in *Nature Communications*, found that wildfire-specific $PM_{2.5}$ was associated with up to ten times the increase in respiratory hospital admissions compared to equivalent concentrations of non-wildfire $PM_{2.5}$. This was a useful starting point for us, because it pushed back against the reflex to aggregate all particulate matter into a single exposure category. Wildfire smoke is chemically and toxicologically distinct, and modeling it separately is not just a methodological nicety; it is scientifically appropriate.

Reid et al. (2016) offer one of the most cited reviews in this space, synthesizing evidence across dozens of studies to conclude that wildfire smoke exposure is consistently associated with respiratory morbidity, with weaker but suggestive evidence for cardiovascular effects. What struck us in this review, and what shaped our thinking, is how much of the evidence base comes from short-term, event-level studies. The question of what cumulative annual exposure does to mortality has been much harder to answer.

That gap has started to close. Doubleday et al. (2020) used a case-crossover design to show that wildfire smoke days in Washington State were associated with a small but statistically detectable increase in all-cause and respiratory mortality. Liu et al. (2017) extended similar logic across 561 U.S. counties and found consistent associations between wildfire smoke days and hospital admissions among older adults. Then came Ma et al. (2024), which is arguably the most methodologically ambitious study in this space to date: using county-month panel data and a 12-month moving average of wildfire $PM_{2.5}$, they estimate that long-term smoke exposure is associated with measurable excess mortality across the contiguous United States, and crucially, that the smoke–mortality association is stronger in hotter months.

We spent a lot of time with the Ma et al. (2024) paper, because at first glance, it looks close to what we are doing. But there is an important distinction we want to flag. In their setup, heat enters as an effect modifier: they interact smoke $PM_{2.5}$ with a count of heat days and interpret the interaction term. This is informative, but it answers a slightly different question than ours. They are asking, "Does heat amplify the effect of chronic smoke exposure?" We are asking, "What is the specific effect of days on which heat and smoke occur simultaneously, beyond what either exposure would predict on its own?" The difference is subtle, but the policy implication is not. If the Ma et al. (2024) interaction is driven mostly by smoke becoming more dangerous in hotter seasons, then the intervention would look like year-round smoke reduction. If our compound-day effect is large, then the intervention looks more like targeted, short-horizon protection (e.g., cooling centers paired with HEPA filtration) on days when both hazards are forecast together. Our paper, in a sense, seeks to isolate the action that Ma et al. (2024) absorbed into a cross-product term.

Any credible study of compound heat–smoke exposure has to take the enormous literature on heat mortality seriously, because otherwise we risk attributing to "compound" effects what is really just heat doing what heat does. Basu (2009) remains one of the cleanest reviews of this literature, summarizing decades of evidence that high ambient temperatures increase all-cause and cause-specific mortality, particularly from cardiovascular and respiratory causes. Gasparrini et al. (2015), in a landmark *Lancet* study spanning 384 locations across 13 countries, estimate that roughly 7% of all deaths in the studied populations are attributable to non-optimal temperature, with both heat and cold contributing.

What we found useful in the Gasparrini et al. (2015) paper, methodologically, is the way it distinguishes between moderate and extreme heat. Much of the mortality burden comes from moderately warm days, not just the tails. That nuance matters for how we construct our heat indicator, which, following Hu et al. (2024), uses a warm-season percentile threshold rather than an absolute temperature cutoff. A 95°F day in Phoenix does not carry the same physiological meaning as a 95°F day in Seattle, and using location-specific percentiles keeps the indicator comparable across counties with very different baseline climates.

From an economic perspective, Deschênes & Greenstone (2011) and Barreca et al. (2016) show that heat-related mortality in the U.S. has declined substantially over the twentieth century thanks to air conditioning and broader adaptation, but that climate projections still imply meaningful excess mortality under business-as-usual scenarios. Barreca et al. (2016) is particularly useful for us, because it uses a panel-data design conceptually similar to ours, exploiting within-county variation in temperature over time to identify effects. Reading their paper gave us some methodological reassurance that a county fixed-effects approach is defensible in this setting, and it shaped how we think about the assumptions we need to check later in the analysis.

2.3 When Heat and Smoke Collide: The Compound Climate Hazards

Here is where the literature is still thin, and here is where we hope to contribute. The idea that climate hazards can interact rather than simply co-occur has been sharpened in recent years by Zscheischler et al. (2018) in *Nature Climate Change*, who argue that "compound events" constitute a distinct category of climate risk that cannot be reduced to the sum of their components. Their framework pushed us to carefully consider whether the health effect of heat-plus-smoke is genuinely more than the sum of its parts, or whether it only appears that way because heat and smoke are correlated in time and space. This is, in effect, the entire identification challenge of our paper compressed into a single conceptual question.

Empirically, a handful of papers have begun to document the geography of compound exposure. Walker et al. (2025) map co-occurring wildfire smoke and extreme heat across the western U.S. from 2006 to 2020, and Jones-Ngo et al. (2025) do something similar for California specifically, showing that compound exposure days have increased markedly and are unequally distributed across racial and socioeconomic lines. These papers are indispensable for us, both because they define the exposure metric we use and because they make the environmental-justice stakes of the question very clear. But they are primarily descriptive. Neither paper attempts to estimate a causal effect of compound exposure on a health outcome, and that is essentially the baton we are trying to pick up.

On the mechanism side, there is a strand of European epidemiology that has studied air pollution–temperature interactions well before the wildfire smoke literature caught up. Stafoggia et al. (2008) and later Analitis et al. (2018) examined how ambient $PM_{2.5}$ (not specifically from fires) interacted with high temperatures in European cities, and both found consistent evidence of synergistic effects on mortality, meaning the combined effect exceeds what would be predicted from adding the two effects separately. The biological rationale is intuitive: heat stress increases respiratory and cardiovascular demands, which, in turn, makes the body more vulnerable to particulate-driven inflammation. If this synergy hypothesis carries over from general-urban $PM_{2.5}$ to wildfire-specific $PM_{2.5}$, we would expect to see a positive coefficient on compound days in our model even after controlling for heat-only days and smoke-only days separately. That is, very literally, the prediction we are testing.

3 Data

3.1 Data Collection[1]

The data is collected manually across multiple datasets. For climate and exposures, wildfire smoke and compound days are based on widely used satellite- and model-derived products. Daily wildfire smoke $PM_{2.5}$ comes from the Childs et al. (2022) dataset, which uses machine learning on ground monitors, satellite AOD, and reanalysis to produce 10-km daily smoke $PM_{2.5}$ estimates across the contiguous US from 2006–2020 and has been shown to perform

[1] The data and variable variable codebook is accessible through the link: `https://docs.google.com/spreadsheets/d/11DSN9iIxrBU5Q5veg9CkoVeWOXAjbCcpphCzh0HDvI4/edit?usp=sharing`

well in out-of-sample prediction. Extreme heat and co-exposure days come from Hu et al. (2024)'s tract-level dataset for 11 Western states, which combines high-resolution temperature data with smoke and burn-zone information and has been used specifically to quantify EH–WFS co-exposure over 2006–2020. These sources provide consistent, spatially resolved daily series that I will aggregate to county–year counts, so measurement error is mostly classical (e.g., some misclassification of smoke or heat days) rather than systematic undercoverage, although there is always some uncertainty in satellite/model products, especially in complex terrain.

The county-level mortality rates and population come from Nassikas (2024), which are generally complete but use suppression for very small counts (fewer than 10 deaths), leading to censoring in sparsely populated counties or for cause-specific mortality. However, the effect is minimal. When processing the data, there are 4 counties where their mortality rate is flagged as suppressed, reducing the counties' sample size from 414 to 410 counties.

Socioeconomic and demographic variables (poverty, race/ethnicity, age) are drawn from ACS and Census, which are standard, high-quality sources at the county level, but many measures are based on 5-year pooled estimates and must be interpolated across years, introducing some temporal smoothing.

3.2 Variables

The treatment (T) is defined as annual compound heat–smoke exposure at the county–year level for 410 counties in eleven Western U.S. states from 2006 to 2020. Conceptually, it measures the number of days in a given year in which a county experiences extreme heat and wildfire smoke simultaneously. We construct this variable using Hu et al. (2024)'s Western U.S. tract–day dataset, which provides binary indicators for extreme heat (EH) and wildfire smoke (WFS) based on predefined thresholds (e.g., temperature above a warm-season percentile cutoff for EH and satellite- or model-based smoke exposure for WFS). We first aggregate tract-level indicators to the county–day level using population or area weights, generating binary indicators for whether county i experiences a heat day or a smoke day on day d in year t. A compound day is defined as any county–day in which both indicators equal one. The treatment is then the annual count of such compound days within county i and year t. This produces a discrete, quantitative variable on a ratio scale, where zero indicates no compound exposure and both differences and ratios are meaningful; empirically, values are expected to range from zero to several dozen days per year.

The outcome (Y) is the annual all-cause mortality rate at the county–year level for the same spatial and temporal units. Let $deaths_{it}$ denote the total number of deaths in county i during year t, and $population_{it}$ is the corresponding mid-year population. The outcome is defined as

$$\frac{deaths_{it}}{population_{it}} \times 100{,}000,$$

expressed as deaths per 100,000 residents per year. This is a continuous, quantitative variable on a ratio scale: zero indicates no deaths, and both absolute differences and proportional comparisons of mortality rates are interpretable.

The confounding variables include social and climate confounders. Social confounders include socioeconomic status, demographics, and population density. Socioeconomic and demographic vulnerability variables function as core confounders because they plausibly influence both exposure patterns and mortality. Socioeconomic status (proxied by poverty rate and population density) affects housing quality, access to air conditioning and filtration, occupational exposure, and residential sorting into higher-risk areas, thereby influencing Smoke and Heat and the likelihood of overlap. Population density belongs in the graph for two reasons: denser counties may experience different exposure dynamics (e.g., urban heat island effects increasing Heat days, or different smoke dispersion patterns), and density strongly correlates with healthcare access, infrastructure, and baseline mortality rates. Demographic composition (e.g., percentage over age 65 and racial/ethnic composition) similarly affects mortality directly through age structure and differential health vulnerability, while also influencing historical residential segregation, land-use policy, and economic inequality that may affect both T and Y.

For climate confounders, yearly temperature and precipitation are structural climatic characteristics that shape the frequency of extreme heat and wildfire smoke exposure across counties, and are therefore included as baseline controls to account for shared climatic drivers of both exposure and mortality.

3.3 Summary Statistics

The summary statistics for the county-year panel dataset covering 400 counties across 11 western US states from 2006 to 2020 (N = 5,810) are shown in Table 1. The mean all-cause mortality rate is 916.66 per 100,000 population, ranging from 135.32 to 2,857.14, reflecting substantial variation across counties. Compound wildfire exposure is highly right-skewed, with a mean of 97.91 days but a median of only 9 days. Thus, we introduced a log transformation, converting compound days to log compound days, which was used in the regression model (mean = 2.45, SD = 1.98). Among the demographic controls, the average county has 17% of its population aged 65 and over, 64% in the working-age population, and 19% children. The average poverty rate is 15.01% (SD = 5.51), with considerable variation ranging from 2.4% to 53%. The mean temperature is 9.47°C, and the precipitation is 626.93 mm. They range from -0.08°C to 23.62°C for temperature and from 18.26 mm to 3557.32 mm for precipitation, reflecting the diverse climate conditions across the western US.

Table 1: Summary Statistics

Variable	N	Mean	SD	Min	Median	Max
Mortality rate	5810	916.66	306.24	135.32	890.58	2857.14
Compound days	5810	97.91	548.98	0.00	9.00	25577.00
Log compound days	5810	2.45	1.98	0.00	2.30	10.15
Heat days	5810	342.68	1696.26	0.00	33.00	45187.00
Hot days	5810	45.86	36.68	0.00	37.65	210.10
Smoke days	5810	1235.47	4141.07	0.00	274.50	164364.00
Age 65+	5810	0.17	0.06	0.04	0.16	0.43
Working age 15–64	5810	0.64	0.04	0.45	0.64	0.80
Children 0–14	5810	0.19	0.04	0.09	0.19	0.32
White pop ratio	5810	0.90	0.11	0.17	0.94	0.99
Log pop density	5810	2.99	1.77	0.25	2.64	9.85
Poverty rate	5810	15.01	5.51	2.40	14.30	53.00
Mean temperature	5810	9.47	4.02	-0.08	9.04	23.62
Precipitation	5810	626.93	524.61	18.26	463.22	3557.32

3.4 Exploratory Data Analysis (EDA)

See the full Explanatory Analaysis in Appendix C. The exploratory analysis reveals three key patterns. First, compound exposure is highly skewed and unevenly distributed, requiring transformation for meaningful analysis. Second, the raw relationship between exposure and mortality appears weak and noisy, suggesting that simple correlations are insufficient to capture the underlying relationship. Third, both exposure and mortality are correlated with demographic and environmental factors such as age structure, poverty, population density, and temperature.

These findings highlight the presence of potential confounding, where factors influencing exposure may also independently affect mortality outcomes. As a result, the observed associations in the raw data cannot be interpreted causally. Instead, a structured framework is required to clearly distinguish between direct effects and confounding pathways.

To address this, we next introduce a causal diagram (DAG) that formalizes the relationships between exposure, mortality, and relevant confounders, and guides the selection of control variables in the empirical strategy.

A key limitation of the TWFE specification is the potential presence of unobserved, time-varying county-level confounders that are not fully absorbed by fixed effects or observed climate controls. In particular, factors such as local public health interventions (e.g., heat warning systems, wildfire response measures) and behavioral adaptation (e.g., increased air conditioning use, reduced outdoor activity during extreme events) may evolve over time within counties and respond to anticipated or realized exposure conditions. These factors plausibly affect both realized exposure (through mitigation or avoidance) and mortality risk, thereby opening backdoor paths that are not blocked by county- or year-fixed effects. Because such variables are difficult to measure consistently at high temporal and spatial resolution, they are not included in the baseline model. As a result, the estimated effects should be interpreted with caution, as they may partially reflect residual confounding from adaptive responses or policy changes that are correlated with exposure dynamics.

4 Methodology

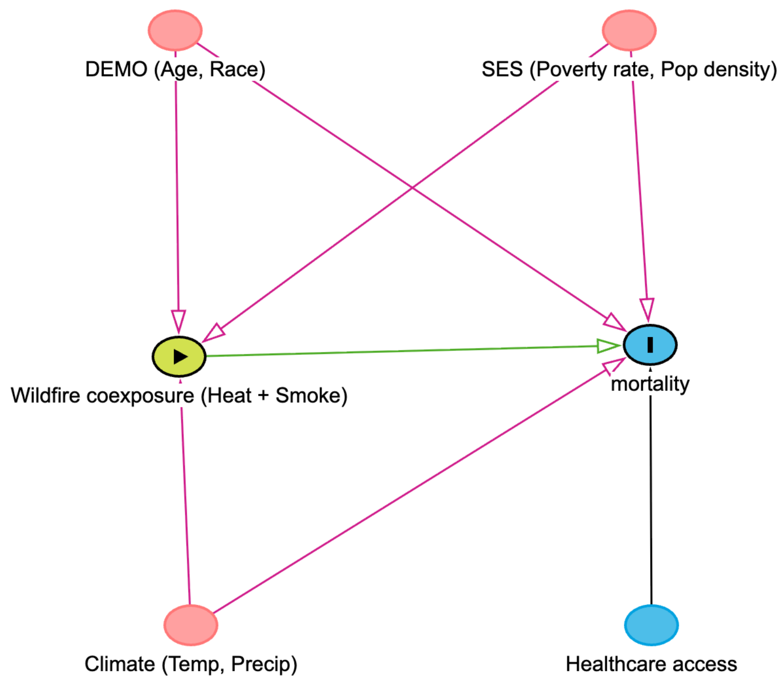
[2]

4.1 Causal diagram

In the simple regression model of mortality rate (Y) on the number of compound heat–smoke days (T) would be badly confounded because the counties that experience more compound heat–smoke exposure are systematically different from those that experience less, in ways that also affect mortality.

[2] The codebook is accessible through the link: `https://colab.research.google.com/drive/1ge7Dpt-WgkB06QLIH7XzBeFtmKWYsYj8?usp=sharing`

Figure 1: Directed Acyclic Graph (DAG)



At least five channels from the DAG (Figure 1) that would bias a naive regression are:

1. **Poverty rate** Poorer, more socioeconomically vulnerable counties have higher baseline mortality and are often more exposed to smoke and heat (e.g., housing quality, AC access, occupational exposure). This opens backdoor paths like $T \leftarrow \text{Poverty} \rightarrow Y$ and $T \leftarrow \text{Smoke} \leftarrow \text{Poverty} \rightarrow Y$, so higher compound days are mechanically correlated with higher mortality even absent a causal effect of T on Y.

2. **Population density (as proxy for urbanity).** Urban counties differ from rural ones in both exposures and health: they may have more heat island effects, different smoke dispersion, and different healthcare access and baseline mortality. Paths like $T \leftarrow \text{URBAN} \rightarrow Y$ and $T \leftarrow \text{Heat} \leftarrow \text{URBAN} \rightarrow Y$ mean URBAN simultaneously pushes up compound days and mortality, biasing the regression.

3. **Age structure (AGE).** Counties with older populations have higher mortality rates regardless of environmental exposure and may be located in regions with particular

smoke/heat patterns. The path $T \leftarrow \text{AGE} \rightarrow Y$ generates confounding: counties with more compound days may also have more older residents, inflating the apparent effect of T on Y.

4. **Racial/ethnic composition (RACE).** Racialized exposure and vulnerability (e.g., residential segregation, occupational structure) affect both where smoke/heat occurs and who dies from them. Paths like $T \leftarrow \text{RACE} \rightarrow Y$ and mixed paths via Smoke/Heat mean that differences in compound days correlate with differences in structural health risk.

5. **Underlying climate and land conditions (TEMP, PRECIP via Smoke and Heat).** Hotter, drier, and more fire-prone climates (and specific land cover types) simultaneously increase Smoke and Heat days and can affect mortality through other mechanisms (e.g., agriculture, chronic heat stress). This shows up in paths like $T \leftarrow \text{Smoke} \leftarrow \text{TEMP} \rightarrow \text{Heat} \rightarrow Y$ and $T \leftarrow \text{Heat} \leftarrow \text{PRECIP} \rightarrow \text{Smoke} \rightarrow Y$, where climate variables create spurious correlation between compound days and mortality.

6. **Healthcare access** We assume that Healthcare access only affect Mortality rate but not wildfire coexposure, while keeping poverty rate and population density constant. Therefore, it is not in the least of minimum required variable to block all backdoor path.

Because all of these factors shift both T and Y, a simple regression $Y_{it} = \alpha + \beta T_{it} + \varepsilon_{it}$ would attribute their joint influence to β, overstating or understating the true causal effect of compound exposure on mortality.

4.2 Quasi-experimental model: TWFE

A quasi-experimental approach, specifically the two-way fixed effects (TWFE) design, would yield a less biased estimate, as TWFE exploits within-county and year-to-year changes in compound exposure. The county fixed effects absorb all stable county characteristics that could simultaneously drive exposure and mortality, and the year fixed effects absorb shocks that affect all counties in a given year, such as a nationwide flu outbreak or a federal policy change.

Data structure and estimand. We analyze county-by-year panels in which the outcome is an annual mortality measure (count or rate) constructed for each county-year. The estimand of primary interest is the **within-county effect of wildfire coexposure on mortality**: specifically, the average change in the outcome associated with a one-unit within-county change in the exposure measure, conditional on observed time-varying covariates and common time shocks.

Primary estimation strategy. The core econometric framework is the linear two-way fixed-effects model. The baseline specification is:

$$y_{it} = \beta E_{it} + X_{it}'\gamma + \alpha_i + \lambda_t + \varepsilon_{it}, \tag{1}$$

where y_{it} is the outcome for county i in year t, E_{it} is an exposure measure, X_{it} is a vector of time-varying controls (including yearly average temperature, yearly sum precipitation, and yearly poverty rate), α_i denotes county fixed effects and λ_t denotes year fixed effects.

Modification of DAG for TWFE Model

Climate controls In a TWFE setup where county FE and year FE already capture time-invariant trends, climate controls matter because the fixed effects only remove *average* differences across units and common shocks over time, not idiosyncratic weather variation that co-moves with both treatment exposure and outcomes. For example, suppose your outcome is mortality and your treatment is wildfire smoke exposure. In a given year, unusually hot and dry conditions can both (i) increase wildfire intensity and smoke dispersion and (ii) directly raise mortality through heat stress and cardiovascular strain. If you omit temperature, the model may incorrectly attribute mortality changes driven by heat waves to wildfire exposure. Similarly, precipitation shocks can matter in agricultural or socioeconomic contexts: drought conditions may increase poverty or displacement while also elevating fire risk, creating a spurious correlation between "treatment" (fire exposure) and outcomes that is actually driven by underlying climate anomalies. Climate controls therefore help isolate the treatment variation that is not simply reflecting contemporaneous weather conditions that independently affect the outcome.

Socioeconomics and demographic confounders However, we exclude time-varying socioeconomic covariates (age structure, racial composition, and log population density) from the baseline TWFE specification for both empirical and causal identification reasons. First, these variables exhibit extremely limited within-county variation over time, as shown by the variance decomposition (ranging from approximately 0.0005 (log population density) to less than 0.05 for most demographic shares) (see Appendix A). In a TWFE design, identification is driven entirely by within-unit variation, so near-time-invariant covariates contribute little additional confounding control beyond what is already absorbed by county fixed effects, while potentially increasing estimation noise due to small and noisy within-county fluctuations.

Second, from a causal perspective, these variables are not clearly pre-determined confounders in a short-run climate–mortality setting. Consider a migration channel: an extreme heat event increases local temperature exposure, which raises health risks and induces out-migration among vulnerable populations (e.g., elderly or low-income individuals) while attracting in-migration of relatively healthier or higher-income individuals with greater adaptive capacity (e.g., access to air conditioning or flexible work). This compositional shift changes observed demographic structure such as age distribution, income, and baseline health, within the population at risk. In turn, these changes mechanically affect measured mortality rates, independently of the direct physiological effect of heat. The resulting causal path can be summarized as:

climate exposure $\rightarrow$ migration / residential sorting $\rightarrow$ demographic composition $\rightarrow$ mortality.

Because demographic variables lie on this pathway, conditioning on them would absorb part of the total effect of climate exposure on mortality, introducing post-treatment bias. In this setting, demographic composition is a "bad control" in the sense of blocking a mediator rather than isolating exogenous variation in exposure. Therefore, these covariates are excluded from the main specification and instead used in robustness checks and sensitivity analyses.

Poverty rate warrants separate consideration because, unlike other demographic variables, it can still function as a time-varying confounder in a TWFE setting. Even after accounting for county and year fixed effects, short-run fluctuations in local economic conditions may simultaneously influence both exposure (e.g., emissions, activity patterns, or vulnerability to compound events) and mortality risk (through healthcare access, baseline health, or resilience). These within-county changes are not fully absorbed by fixed effects and therefore remain a potential source of confounding in the identifying variation. At the same time, poverty may also be partially endogenous to local conditions, making its causal role ambiguous. By comparing a model that adds poverty rate to the baseline specification (defined as mortality on co-exposure, temperature, precipitation, county fixed effects, and year fixed effects) we can assess whether residual within-county, time-varying economic conditions confound the estimated relationship between exposure and mortality. If the coefficient on co-exposure remains stable after including poverty rate, this suggests that short-run socioeconomic fluctuations are not driving the results. Conversely, meaningful changes in the estimate would indicate that time-varying economic conditions, not fully absorbed by fixed effects, play a confounding role in the identifying variation.

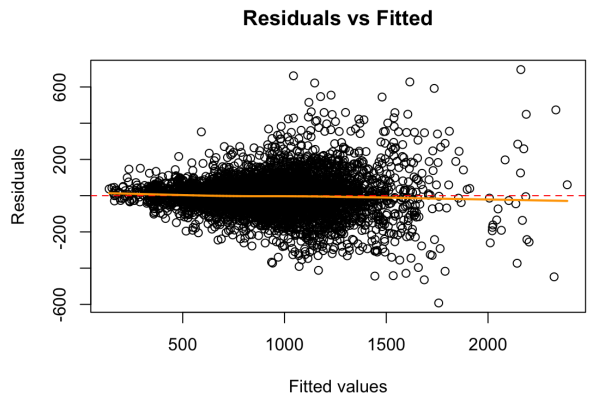
4.3 Assumptions

For assumptions, we evaluate linearity, homoscedasticity, no autocorrelation, normal errors, multicollinearity, and exogeneity. While panel data often violate heteroscedasticity, autocorrelation, and multicollinearity, we believe that conducting these tests remains essential for diagnosing model adequacy and establishing a baseline (Juliana et al., 2025).

To test the linearity and homoscedasticity assumptions, we plotted the regression model residuals against their predicted values (Figure 2). The linearity assumption is satisfied: the residuals are centered around $y = 0$ across the full range of fitted values, as shown by the orange line, which is almost flat, indicating no curve, bend, or systematic pattern. Thus, we can rely on the model's coefficients and standard errors.

Regarding the homoscedasticity assumption, the spread of residuals is not constant across fitted values: at the low fitted values, from approximately 300 to 500, it is a tight and narrow spread; at mid fitted values, approximately from 800 to 1500, the spread starts to expand in both directions; at high fitted values, beyond 1500, there are fewer observations but the spread has become wider. This fan-like shape indicates heteroscedasticity, meaning the output's standard errors cannot be relied upon.

Figure 2: Scatter Plot of the Residuals against their Predicted Values



Regarding the no autocorrelation assumption, we conducted the Breusch–Godfrey test using the `pbgtest` function in R for panel data. With the significantly low p-value ($p < 0.001$), we reject H_0 at any conventional significance level. There is strong statistical evidence of serial correlation in the idiosyncratic errors in the panel data. Although the fixed effects absorb stable county-level differences and common-year shocks, within-county serial correlation persists in the idiosyncratic errors. With autocorrelation, the standard errors cannot be relied upon.

Table 2: Breusch–Godfrey/Wooldridge Test

Test	Statistics (Chi-sq)	DF	P-value
Breusch–Godfrey/Wooldridge	26.533	1	2.59×10^{-7}

Regarding the normal error assumptions, while the histogram follows a bell-shaped, normal distribution around zero, the quantile-quantile (Q–Q) plot displays an S-curve line (Figure 3 and 4). Even though the middle section around the theoretical quantiles from -1 to 1 follows the normal distribution, the left and right tails are off from the normal distribution line. Additionally, we conducted the Anderson–Darling test, which yielded a statistic of 123.59 and a p-value of 2.2×10^{-16}, which is almost 0. The large statistic value indicates strong departure from normality, and a low p-value rejects the null hypothesis that the residuals are not normally distributed. Violating the normal error assumptions means the standard error in the output is unreliable. Nevertheless, given $n = 5{,}810$ observations, we could say the normal error assumption is satisfied under the Central Limit Theorem.

Figure 3: Histogram of the Residuals

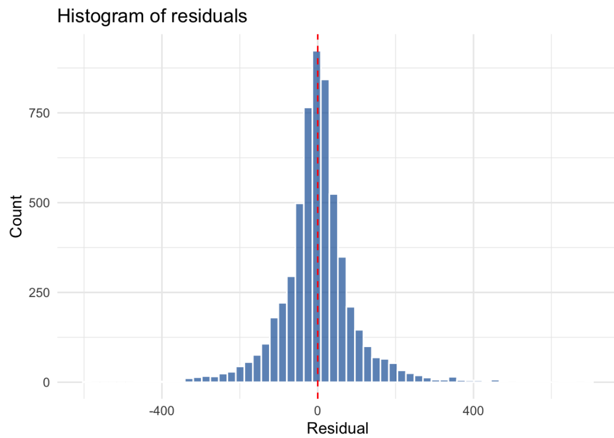


Figure 4: Quantile-Quantile (Q-Q) Plot

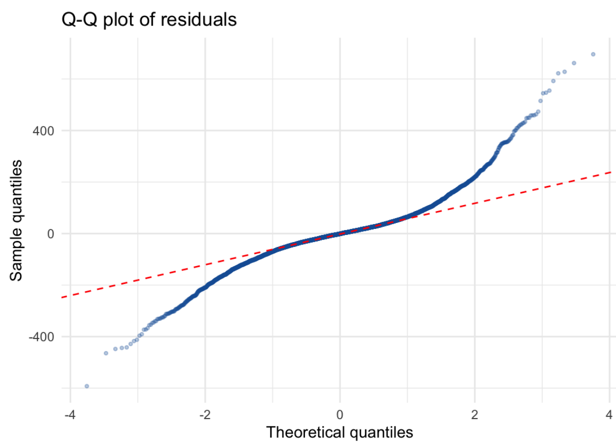


Table 3: Anderson–Darling Test

Test	Statistics	P-value
Anderson–Darling test	123.59	2.2×10^{-16}

To assess multicollinearity, Variance Inflation Factors (VIFs) were computed for the independent variables (Table 4). All values below 2.2 indicate no multicollinearity. Specifically, `log_compound` and `poverty_rate` show almost no correlation with the other regressors, with VIF values of 1.057 and 1.074, respectively. The `log_pop_density_tv` is also very low (VIF = 1.322), and `pct_age_65_plus` and `w_population_ratio` show low-to-moderate correlations (VIF = 2.113 and 2.191, respectively). The multicollinearity assumption is satisfied, so that the coefficients and standard errors are reliable.

Table 4: Variance Inflation Factors (VIF)

Variable	VIF
log_compound	1.057
pct_age_65_plus	2.113
w_population_ratio	2.191
log_pop_density_tv	1.322
poverty_rate	1.074

Regarding exogeneity, we argue that the assumption is satisfied by controlling for key confounders of the relationship between wildfire exposure and mortality. County fixed effects control for stable differences across counties, such as geography, baseline health infrastructure, and long-run socioeconomic conditions. Year fixed effects control for events affecting all counties in the same year, such as national policy changes or disease outbreaks. We also include time-varying controls for age structure, population composition, population density, and poverty rate, which account for within-county changes that independently affect mortality. The remaining year-to-year variation in wildfire exposure is therefore driven by weather and climate conditions outside county control, satisfying the exogeneity assumption. While unobserved confounders cannot be fully ruled out, our variable selection is consistent with the existing literature on wildfire exposure and health outcomes, and the lead variable test provides additional empirical support for the assumption.

Dynamic diagnostics and pre-trend (placebo) tests. To assess the identifying parallel-trends assumption and the timing of effects, we estimate a dynamic specification that includes two placebo leads and two lags of the exposure:

$$y_{it} = \sum_{k=-2}^{2} \beta_k E_{i,t+k} + X_{it}'\gamma + \alpha_i + \lambda_t + \varepsilon_{it},$$

where $E_{i,t+k}$ is the exposure k periods relative to the contemporaneous year ($k < 0$ are leads). Leads serve as placebo tests: statistically indistinguishable from zero supports no pre-trend in the outcome that anticipates future exposure. We report individual lead/coefficient estimates, a joint Wald test for the null $\beta_{-2} = \beta_{-1} = 0$, and the cumulative contemporaneous + lagged effect $\sum_{k=0}^{2} \beta_k$ with its standard error computed from the model covariance matrix via

$$\mathrm{Var}\left(\sum_{k=0}^{2} \hat{\beta}_k\right) = \sum_{i=0}^{2}\sum_{j=0}^{2} \widehat{\mathrm{Cov}}(\hat{\beta}_i, \hat{\beta}_j).$$

We plot the lead/lag point estimates and 95% CIs against event time (relative time $-2, \ldots, +2$) as the primary diagnostic figure.

Leads/lags results. The lead coefficients (t = 2, 1) are both statistically indistinguishable from zero, and a joint Wald test fails to reject the null that they are jointly zero (p = 0.48).

This provides no evidence of differential pre-trends and supports the identifying assumption that, absent exposure, mortality would have evolved similarly across counties.

5: At impact (t = 0), the estimated effect is positive (1.29) but imprecise and not statistically significant. The subsequent lags (t = 1, 2) are negative but also highly imprecise, with confidence intervals that comfortably include zero. There is no clear evidence of persistence or delayed amplification; instead, the dynamic profile is noisy and does not exhibit a consistent temporal pattern.

Taken together, the results suggest two things. First, the absence of pre-trends strengthens causal interpretation by reducing concerns about anticipatory behavior or omitted confounders. Second, the post-treatment dynamics indicate that any effect of climate exposure on mortality is either small relative to the available statistical power or highly transitory and difficult to distinguish from zero in this sample. The wide confidence intervals across all event times—combined with the very low within R^2—point to limited identifying variation in the dynamic specification, which likely contributes to the imprecision of the estimates.

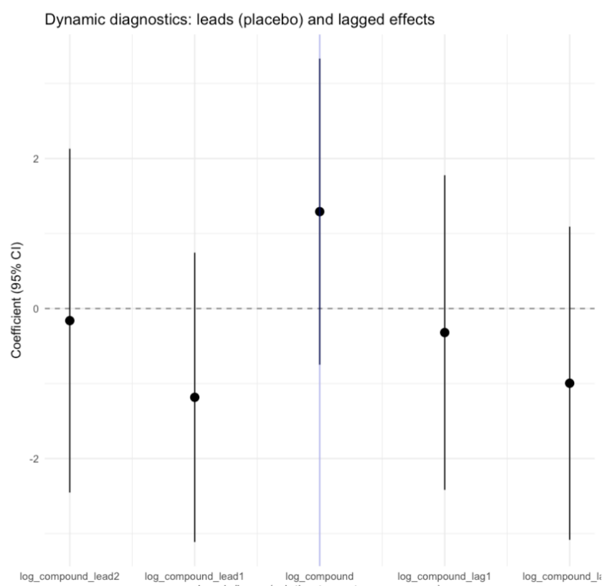


Figure 5: *Event-time (lead–lag) estimates of the effect of climate exposure on mortality.* The figure plots coefficients from a two-way fixed effects model of mortality on contemporaneous, lagged (t+1, t+2), and lead (t1, t2) values of log climate exposure. The omitted reference period is t = 1. County and year fixed effects are included, with standard errors clustered at the county level. Points denote point estimates and vertical bars indicate 95% confidence intervals. Leads serve as placebo tests for pre-trends, while lags capture dynamic adjustment following exposure.

4.4 Model specification

We estimate a series of county-by-year two-way fixed effects (TWFE) models to evaluate the relationship between compound exposure and mortality. All specifications include county fixed effects (α_c) and year fixed effects (δ_t), and standard errors are clustered at the county level.

Baseline specification. Our primary model includes the exposure of interest and key climatic controls:

$$mortality_{ct} = \beta\, log\_compound_{ct} + \gamma_1 ppt_{ct} + \gamma_2 tmean_{ct} + \alpha_c + \delta_t + \varepsilon_{ct}. \tag{2}$$

Socioeconomic adjustment. To assess sensitivity to time-varying economic conditions, we augment the baseline model with poverty rate:

$$mortality_{ct} = \beta\, log\_compound_{ct} + \gamma_1 ppt_{ct} + \gamma_2 tmean_{ct} + \gamma_3 poverty_{ct} + \alpha_c + \delta_t + \varepsilon_{ct}. \tag{3}$$

Extended covariate model. We further estimate a specification including additional demographic and socioeconomic controls:

$$\begin{aligned} mortality_{ct} = {} & \beta\, log\_compound_{ct} + \gamma_1 ppt_{ct} + \gamma_2 tmean_{ct} + \gamma_3 poverty_{ct} \\ & + \gamma_4 log\_pop\_density_{ct} + \gamma_5 pct\_age\_65\_plus_{ct} + \gamma_6 pct\_child_{ct} + \gamma_7 w\_population\_ratio_{ct} \\ & + \alpha_c + \delta_t + \varepsilon_{ct}. \end{aligned} \tag{4}$$

Distributed lag model. To capture delayed effects of exposure, we estimate a distributed lag specification including contemporaneous and lagged exposure terms:

$$mortality_{ct} = \beta_0\, log\_compound_{ct} + \beta_1\, log\_compound_{c,t-1} + \beta_2\, log\_compound_{c,t-2} + \gamma_1 ppt_{ct} + \gamma_2 tmean_{ct} + \alpha_c + \delta_t \tag{5}$$

The cumulative exposure effect is defined as:

$$\beta_{cum} = \beta_0 + \beta_1 + \beta_2. \tag{6}$$

Heterogeneity models. To examine heterogeneity, we estimate separate models for subgroup-specific mortality outcomes:

$$mortality_{g,ct} = \beta_g\, log\_compound_{ct} + \gamma_1 ppt_{ct} + \gamma_2 tmean_{ct} + \alpha_c + \delta_t + \varepsilon_{g,ct}, \tag{7}$$

where g indexes subgroups (age 65+, under 65, White, and non-White populations).

This approach allows the exposure effect to vary flexibly across subpopulations while maintaining a consistent identification strategy. However, subgroup analyses may be sensitive to smaller population sizes, and are therefore interpreted as exploratory.

Placebo specification. As a robustness check, we implement a permutation-based placebo test by randomly reassigning exposure time series across counties and re-estimating the distributed lag model. This preserves temporal structure while breaking any true exposure–outcome relationship.

All models include county and year fixed effects and cluster standard errors at the county level.

5 Results

Table 5: Summary of TWFE Estimates (within-county effects)

Panel	Model Description	Term	Estimate	SE	p-value
Baseline Models					
	Baseline TWFE (climate controls)	log_compound	1.239	0.854	0.147
	Baseline + poverty rate	log_compound	1.071	0.855	0.211
Extended Controls					
	Full controls (SES + demographics)	log_compound	1.210	0.795	0.129
Dynamic Models					
	Distributed lag (t, t-1, t-2)	log_compound (t)	1.850	0.899	0.040
	Distributed lag (exclude 2020)	log_compound (t)	1.940	0.825	0.019
	Cumulative effect (t to t-2)	sum of lags	2.916	–	–
Placebo Test					
	Permutation test (cumulative effect)	p-value	–	–	0.244
Heterogeneity (Subgroup Outcomes)					
	Age 65+ mortality	log_compound	-4.546	4.590	0.322
	Age <65 mortality	log_compound	0.902	0.673	0.180
	White mortality	log_compound	1.309	0.916	0.153
	Non-White mortality	log_compound	3.139	1.196	0.518

5.1 Baseline Estimates

Baseline TWFE estimates indicate a positive but statistically imprecise association between compound exposure and mortality. The coefficient on log *compound* is approximately 1.24 (SE = 0.85, $p = 0.15$), and confidence intervals include zero. Climate controls (temperature and precipitation) are not statistically significant in these specifications.

Including poverty rate as an additional control does not materially change the estimated effect (coefficient = 1.07, SE = 0.85), suggesting that short-run socioeconomic variation does not drive the main results. Although poverty rate itself is statistically significant, its inclusion does not alter the exposure coefficient, consistent with limited confounding from time-varying economic conditions.

Adding a full set of socioeconomic controls produces similar conclusions: estimates remain imprecise and close to zero, reinforcing that identification is not driven by observable socioeconomic variation.

5.2 Dynamic Effects

The distributed lag model suggests a modest contemporaneous effect of exposure (coefficient = 1.85, SE = 0.90, $p = 0.04$), with smaller and statistically insignificant lagged effects. The cumulative effect across current and lagged exposure is approximately 2.92, but remains imprecise.

Excluding 2020 yields similar results, with a slightly larger contemporaneous coefficient (1.94, SE = 0.83, $p = 0.02$), indicating that results are not driven by pandemic-related mortality shocks.

5.3 Placebo Test

Permutation-based placebo tests further support the null findings. The empirical p-value is approximately 0.24, indicating that the observed cumulative effect is not distinguishable from random assignments of exposure. This suggests that the estimated effects are not robustly different from noise under the null.

5.4 Heterogeneity by Age and Race

For age-based mortality outcomes, the estimated effect is negative for the 65+ population (−4.55, SE = 4.59) and positive for the under-65 population (0.90, SE = 0.67), with neither estimate statistically distinguishable from zero. If ignoring the statistically insignificant result, the negative coefficient for the 65+ population could reflect potential model misspecification (functional form limitations, temporal aggregation, or compositional and measurement-related artifacts) rather than a true protective effect.

Race-based heterogeneity exhibits larger point estimates for non-White mortality (3.14, SE = 1.20), compared to White mortality (1.31, SE = 0.92). However, both estimates are imprecisely estimated and not robustly significant across specifications, limiting strong inference about differential impacts by race. While the direction of estimates is consistent with potentially higher vulnerability among non-White populations, the wide confidence intervals indicate substantial uncertainty.

These patterns may reflect differences in vulnerability, but could also arise from measurement error, differences in population denominators, or limited statistical power within subgroup outcomes.

6 Robustness Checks

6.1 Robustness of Functional Form

A central concern in the analysis is the large share of zero exposure days and the resulting sensitivity of functional form choices for the compound exposure measure. Approximately 23% of county-year log compound have zero exposure, and a small number of counties exhibit persistently zero exposure throughout the sample. This motivates a series of robustness checks examining whether the main results are driven by the treatment of zero outcomes or the functional transformation of exposure intensity.

First, results are robust to alternative transformations of exposure. Using both log(1 + compound days) and asinh(·) yields positive and statistically significant estimates of similar magnitude to the baseline specification, suggesting that results are not driven by the log transformation or handling of zeros.

Second, we examine whether the effect is driven by the extensive or intensive margin. A two-part model shows that the effect is concentrated in the intensive margin: conditional on exposure, higher compound intensity is positively associated with mortality, while the binary indicator for any exposure is not statistically significant in the full sample.

Third, we explore non-parametric heterogeneity across exposure levels using decile bins. The relationship is broadly monotonic, with higher exposure deciles exhibiting larger coefficients, consistent with an intensity-driven dose-response pattern rather than threshold effects.

Finally, alternative specifications separating positive-only counties and interaction-based models yield consistent positive estimates for the intensive margin, though precision declines when restricting to exposed units only. Overall, robustness checks support the interpretation that the main results are driven by exposure intensity rather than model specification or functional form choices.

See the detailed robustness check results in Appendix B.

7 Conclusion

The paper examined the causal effect of the annual number of days on which extreme heat and wildfire smoke occur simultaneously on county-level mortality rates, controlling for overall heat and smoke exposure. Using a two-way fixed effects framework, the results indicate a positive but statistically imprecise association between compound exposure and mortality.

Across specifications, the estimated coefficient on log compound days ranges from approximately 1.07 to 1.94. In real world impact, a 1% increase in compound exposure (days with both extreme heat-and-smoke), for example, from 20 to 20.2 days per year, is associated with an increase of approximately 0.011 to 0.019 deaths per 100,000 population annually (or 11-19 deaths per 1 million population).

While the direction of the effect is consistent with the alternative hypothesis, these estimates are not statistically distinguishable from zero. Moreover, the implied standardized effect size is very small ($d \approx 0.012$–0.013), indicating that even meaningful changes in exposure correspond to only minor shifts in mortality relative to its overall variation.

Taken together, these results suggest that any average short-run effect of compound heat and smoke exposure at the county-year level is modest in magnitude and difficult to detect with precision. This may reflect behavioral adaptation, avoidance responses, or aggregation effects that attenuate impacts occurring at finer temporal or spatial scales.

The heterogeneity analysis provides suggestive—though statistically imprecise—evidence that impacts may differ across subpopulations, with larger point estimates observed for non-White mortality. While this pattern is consistent with existing evidence on environmental inequality, the lack of statistical precision means these differences cannot be interpreted as causal or robust.

Overall, the findings are most consistent with a setting in which compound exposure does not generate large, detectable average effects on mortality at the county-year level, but may still produce localized or distributional impacts that are masked in aggregate data. Future work using higher-frequency data, more granular exposure measures, or designs with greater statistical power would be necessary to more precisely identify these effects and assess their distribution across populations.

8 Limitations

It is important to recognize that this analysis is subject to several limitations that affect both the identification and interpretation of the estimated effects. First, the empirical strategy relies on county–year level aggregation, which inherently compresses the temporal structure of exposure. Compound heat–smoke events are fundamentally short-run phenomena that occur at the daily level, yet the treatment variable is constructed as an annual count of such days. This aggregation abstracts away from the timing, duration, and intensity of exposure episodes, which may be critical for understanding physiological responses. As a result, the estimated effects reflect an average annual relationship and may understate or obscure short-run nonlinearities in mortality risk associated with concentrated periods of extreme co-exposure.

Second, while the two-way fixed effects framework controls for time-invariant county characteristics and common year shocks, it does not fully address the possibility of unobserved, time-varying confounders. In particular, behavioral and policy responses to climate exposure, such as increased use of air conditioning, wildfire mitigation efforts, or public health interventions, may evolve within counties over time. These adaptive responses are plausibly correlated with both exposure and mortality, and are not directly observed in the data. To the extent that such factors systematically mitigate mortality risk in high-exposure settings, the estimated coefficients may be biased toward zero, limiting the ability to distinguish between true null effects and attenuated causal relationships.

Finally, the construction of the compound exposure measure introduces an additional source of limitation. By defining compound days as the co-occurrence of binary heat and smoke indicators, the analysis implicitly treats all compound exposure days as homogeneous. This approach does not account for variation in intensity, such as differences in temperature extremes or $PM_{2.5}$ concentration levels, nor does it capture potential threshold effects where health impacts may rise disproportionately at higher levels of exposure. Consequently, the treatment variable may mask meaningful heterogeneity in exposure severity, leading to imprecise estimates and limiting the ability to detect stronger effects in extreme conditions.

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A Appendix

Main code notebook: https://colab.research.google.com/drive/1ge7Dpt-WgkB06QLIH7XzBeFtmKWYsYj

Dataset: https://docs.google.com/spreadsheets/d/11DSN9iIxrBU5Q5veg9CkoVeWOXAjbCcpphCzh0HDv

Appendix A: Within-County Variation Across Confounders

Figure 6: Within-County Variance Decomposition

Variable	Total Var	Within Var	Between Var	Within Share	Within SD	Total SD
Log Population Density (tv)	3.131	0.002	3.130	0.0005	0.041	1.770
Working-age Population Share	0.012	0.000	0.012	0.0058	0.008	0.111
Mean Temperature (°C)	16.137	0.431	15.706	0.0267	0.657	4.017
Share Age 0–14	0.002	0.000	0.002	0.0486	0.009	0.040
Precipitation	275219.4	26279.462	248939.935	0.0955	162.109	524.614
Poverty Rate	30.343	3.467	26.877	0.1143	1.862	5.508
Share Age 65+	0.003	0.001	0.003	0.1746	0.024	0.057

A.1 Appendix B: Robustness check

First, we test functional form dependence by re-estimating the model using alternative transformations of compound exposure, including $\log(1+\text{compound days})$ and the inverse hyperbolic sine transformation $\text{asinh}(\cdot)$. These transformations allow us to assess whether the results are sensitive to the treatment of zero values and to alternative assumptions about the nonlinear scaling of exposure intensity.

Second, we separate the exposure process into extensive and intensive components to distinguish between the occurrence of exposure and its severity. The extensive margin is captured by an indicator for whether a county experiences any compound exposure in a given year, which identifies differences between exposed and non-exposed units. Conditional on exposure occurring, the intensive margin is modeled using the log-transformed number of compound days, capturing variation in exposure severity among affected counties. We also estimate a joint specification that includes both terms simultaneously, which allows us to disentangle whether estimated effects are driven by selection into exposure or by dose-dependent responses within exposed observations.

Table 6: Robustness checks: alternative specifications of compound exposure

Model	Specification	Estimate	SE	p-value
Functional form	$\log(1+\text{compound})$	3.347	1.496	0.026
	asinh(compound)	3.393	1.500	0.024
Extensive vs intensive margin	Any exposure (binary)	1.799	4.092	0.661
	Intensive margin (positive sample)	3.347	1.496	0.026

We conduct a series of robustness checks focusing on functional form, exposure structure, dynamic effects, and placebo validity.

First, alternative transformations of exposure yield highly consistent results. Both $\log(1+\text{compound})$ and $\text{asinh}(\cdot)$ specifications produce positive and statistically significant estimates of similar magnitude, indicating that results are not driven by the choice of log transformation or the handling of zero exposure values.

Second, decomposing exposure into extensive and intensive margins shows that the estimated effect is driven primarily by intensity rather than the binary occurrence of exposure. The extensive margin is not statistically significant, while the intensive margin remains positive and significant.

Overall, the robustness analysis supports a consistent interpretation of a positive but intensity-driven relationship between compound exposure and mortality, while highlighting that statistical precision varies across specifications.

A.2 Robustness of Outcome Identification: Binary Outcome

To assess whether the relationship between compound exposure and mortality depends on how the outcome is measured, we construct a binary outcome equal to one if a county-year's mortality rate is above the within-year median and zero otherwise. We then estimate a linear probability model with county and year fixed effects using the same set of controls as in the baseline specification.

The coefficient on log compound exposure is positive and statistically significant ($\beta = 0.0067$, $p = 0.0028$), implying that a one-unit increase in log compound exposure raises the probability of a county-year being in the high-mortality group by approximately 0.67 percentage points. Among controls, the share of elderly population is positively associated with high mortality ($\beta = 1.41$, $p < 0.001$), while population density is negatively associated ($\beta = -0.36$, $p < 0.001$). Poverty rate and population ratio are not statistically significant. Overall, the positive and significant coefficient on compound exposure suggests that the relationship persists even under a binary outcome specification.

A.3 State-Level Clustering

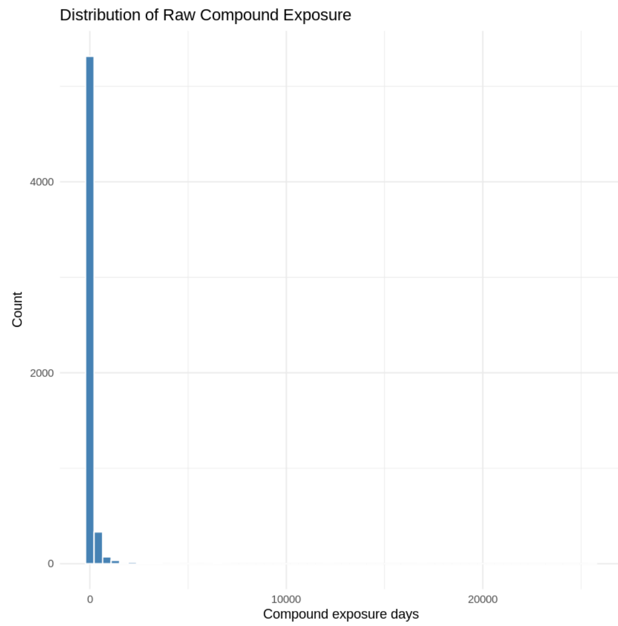
To assess whether the estimated relationship is sensitive to the choice of error clustering, we re-estimate the baseline model using standard errors clustered at the state level instead of the county level. This allows for correlation in unobserved shocks across counties within the same state, capturing broader spatial dependence that may not be fully accounted for in the main specification.

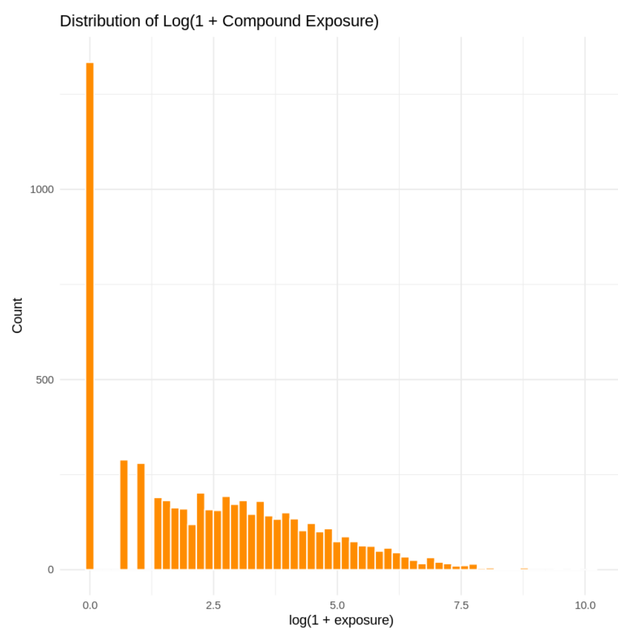
The coefficient on log compound exposure remains unchanged in magnitude ($\beta = 1.14$) when moving from county-level to state-level clustering, indicating that the point estimate is stable across specifications. However, the standard error increases from 0.898 to 1.526, leading to a loss of statistical significance. This suggests that some variation in the data may be correlated across counties within states, reducing the precision of the estimate when broader clustering is allowed. Similarly, the coefficient on poverty remains positive but becomes less precisely estimated. Overall, while the direction and magnitude of the main relationship are preserved, the reduced statistical significance under state-level clustering highlights sensitivity to assumptions about error correlation and reinforces the need for cautious interpretation.

A.4 Appendix C: Explanatory Analysis

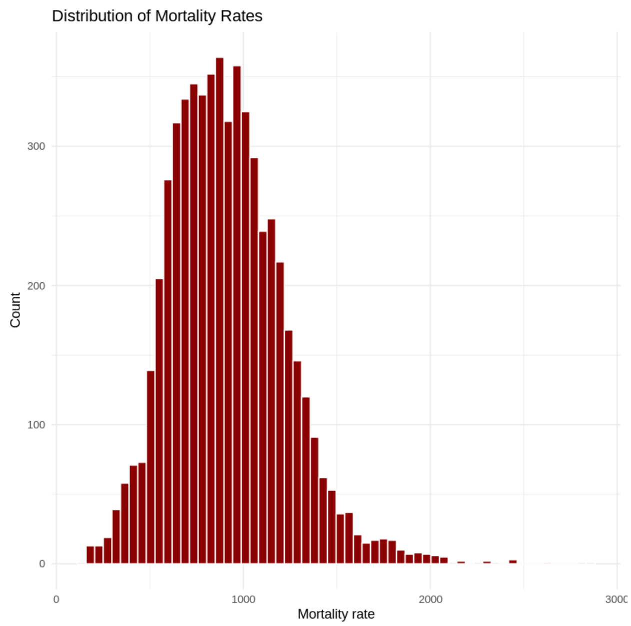
We conduct exploratory data analysis to understand the dataset's structure, the distributions of key variables, and the relationships among exposure, mortality, and potential confounders. This step is important for motivating modeling choices, such as the use of log transformations, fixed effects, and control variables. In particular, the analysis helps assess whether the variables exhibit sufficient variation and whether simple patterns suggest the presence of confounding factors that must be accounted for in the regression framework.

1) Distribution of Compound Exposure. To examine the distribution of wildfire smoke exposure, we plot histograms of both the raw number of compound exposure days and its log-transformed version. Raw compound exposure is highly right-skewed, with most observations near zero and a small number of extreme values. The log transformation reduces skewness and compresses outliers, making the variable more suitable for regression analysis.

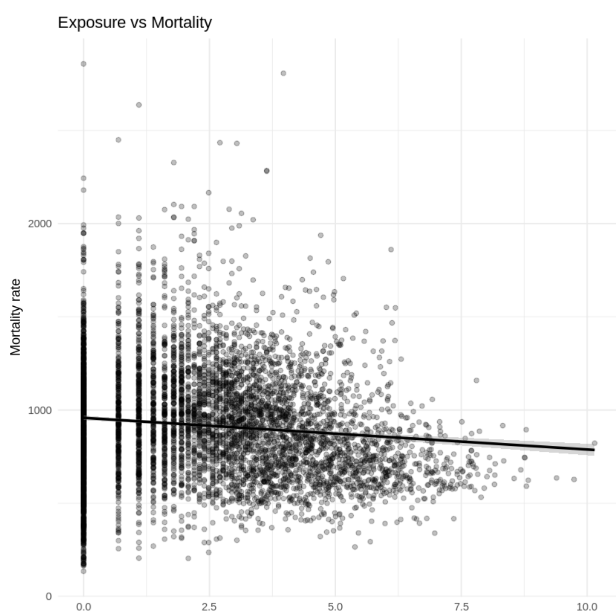




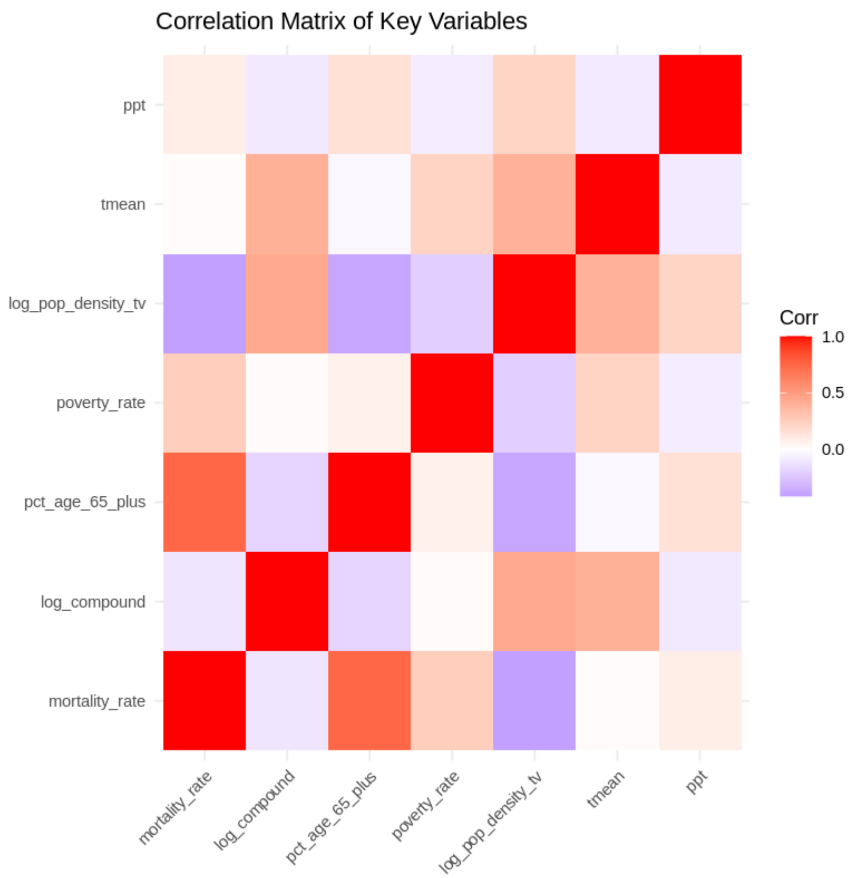
2) Distribution of Mortality Rates. To understand the variation in the outcome variable, we plot the distribution of county-level mortality rates. Mortality rates are moderately dispersed and centered around a stable mean, indicating sufficient variation across counties and years to support statistical analysis.



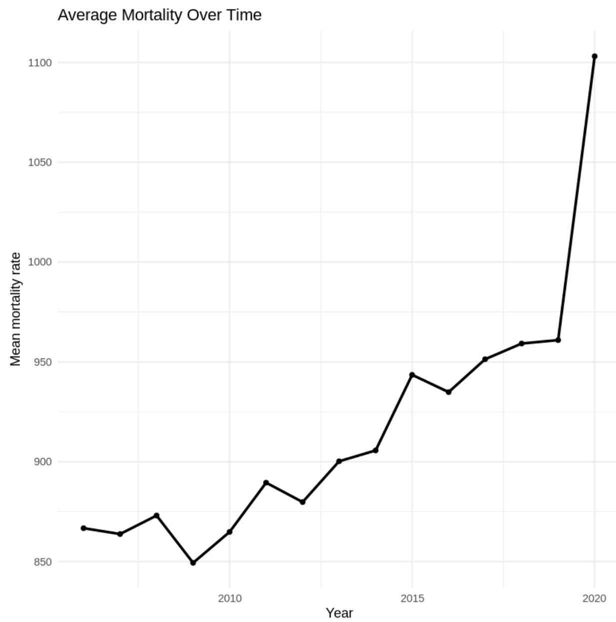
3) Relationship between Exposure and Mortality. To visualize the raw relationship between exposure and mortality, we plot a scatterplot of log compound exposure against mortality rates, along with a fitted linear trend. The relationship between exposure and mortality appears weak and noisy, with no strong visible pattern. This highlights the need for a regression framework that accounts for confounding factors and unobserved heterogeneity.



4) Correlation Between Key Variables. To assess potential confounding, we compute the pairwise correlations between mortality, exposure, and key demographic and environmental variables. Mortality is strongly correlated with age structure and moderately correlated with poverty and population density, while exposure is correlated with density and temperature. These patterns suggest the presence of confounding factors that must be controlled for in the empirical specification.



5) Trends Over Time. To examine aggregate patterns, we plot the average mortality rate over time across all counties. Average mortality exhibits a gradual upward trend over time, indicating the presence of common time effects and motivating the inclusion of year fixed effects in the regression model.


Average Mortality Over Time
Mean mortality rate
1100
1050
1000
950
900
850
2010
2015
2020
Year